

Moeketsi Raselimo

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Professional Experience

Humboldt-Universität zu Berlin

Berlin, Germany

Postdoctoral Research Fellow

Since July 2023

- Leading research in automated software engineering, developing novel methods to improve the reliability, security, and trust of complex software systems.
- Reviewing research papers and software for top-tier software engineering conferences.
- Leading the development of a prototype to find incomplete/bad bug fixes; used the tool to confirm such bad fixes in popular C open-source projects.
- Collaborating on a static analysis tool and paper to find precise bug descriptions.
- Teaching advanced topics in automated software engineering such as automated test case generation and program analysis.
- Supervising and mentoring graduate students.

Stellenbosch University

Stellenbosch, South Africa

Extraordinary (Adjunct) Lecturer

Since September 2024

- Collaborating on a research prototype that learns the structure of the input space of software systems; useful for bug finding and program comprehension.
- Collaborating on grammar-based testing techniques.
- Supervising and mentoring graduate students.

Stellenbosch University

Stellenbosch, South Africa

Postdoctoral Research Fellow

March 2023 - August 2023

- Completed unpublished work from my PhD research.

Education

Stellenbosch University

Stellenbosch, South Africa

PhD in Computer Science

February 2019 - March 2023

- **Specialization:** Program Analysis and Testing
- **Supervisor:** Prof. Bernd Fischer
- **Dissertation:** Fault Localization and Repair for Grammarware

Stellenbosch University

Stellenbosch, South Africa

BSc(Honours) in Computer Science

February 2018 - December 2018

- Graduated **Cum Laude**

Savitribai Phule Pune University

Pune, India

BSc in Computer Science

July 2014 - May 2017

- Graduated with **Distinction**

Publications

1. Martin Eberlein, **Moeketsi Raselimo**, Lars Grunske. 2025. *Which Inputs Trigger my Patch?* In Proceedings of the 2025 IEEE/ACM International Workshop on Automated Program Repair (APR). <https://ieeexplore.ieee.org/document/11029655>.

2. **Moeketsi Raselimo** and Bernd Fischer. 2024. Spectrum-Based *Rule-and Item-Level Localization of Faults in Context-Free Grammars*. Journal of Systems and Software, Volume 215. <https://www.sciencedirect.com/science/article/pii/S0164121224001122>.
3. Dirk Beyer, Lars Grunske, Matthias Kettl, Marian Lingsch-Rosenfeld, and **Moeketsi Raselimo**. 2024. *P3: A Dataset of Partial Program Patches*. In Proceedings of 21st International Conference on Mining Software Repositories (MSR 2024). <https://dl.acm.org/doi/abs/10.1145/3643991.3644889>
4. **Moeketsi Raselimo**, Lars Grunske, and Bernd Fischer. 2023. *Static Test Case Prioritization Strategies for Grammar-Based Testing*. In Proceedings of the 38th IEEE/ACM International Conference on Automated Software Engineering Workshops (ASEW). <https://doi.org/10.1109/ASEW60602.2023.00025>.
5. **Moeketsi Raselimo** and Bernd Fischer. 2023. *Automatic Passive and Active Repair for Grammars*. Technical Report. <https://www.moeketsi.me/assets/pdf/repair-tech-report.pdf>
6. **Moeketsi Raselimo**. 2022. *Fault Localization and Repair for Grammarware*. PhD Thesis. Stellenbosch University. <https://tinyurl.com/moeketsi-thesis>
7. **Moeketsi Raselimo** and Bernd Fischer. 2021. *Automatic Grammar Repair*. In Proceedings of the 14th ACM SIGPLAN International Conference on Software Language Engineering (SLE '21). <https://doi.org/10.1145/3486608.3486910>.
8. Chelsea Barraball, **Moeketsi Raselimo**, and Bernd Fischer. 2020. *An Interactive Feedback System for Grammar Development (Tool Paper)*. In Proceedings of the 13th ACM SIGPLAN International Conference on Software Language Engineering (SLE '20). <https://doi.org/10.1145/3426425.3426935>.
9. Phillip van Heerden, **Moeketsi Raselimo**, Konstantinos Sagonas, and Bernd Fischer. 2020. *Grammar-Based Testing for Little Languages: An Experience Report with Student Compilers*. In Proceedings of the 13th ACM SIGPLAN International Conference on Software Language Engineering (SLE '20). <https://doi.org/10.1145/3426425.3426946>.
10. **Moeketsi Raselimo** and Bernd Fischer. 2019. *Spectrum-Based Fault Localization for Context-Free Grammars*. In Proceedings of the 12th ACM SIGPLAN International Conference on Software Language Engineering (SLE '19). <https://doi.org/10.1145/3357766.3359538>.
11. **Moeketsi Raselimo**, Jan Taljaard, and Bernd Fischer. 2019. *Breaking Parsers: Mutation-Based Generation of Programs with Guaranteed Syntax Errors*. In Proceedings of the 12th ACM SIGPLAN International Conference on Software Language Engineering (SLE '19). <https://doi.org/10.1145/3357766.3359542>. **ACM Distinguished Paper Award**.

Presentations

Automatic Grammar Repair
SLE 2021 Conference

Virtual

Spectrum-Based Fault Localization for Context-Free Grammars
SLE 2019 Conference

Athens, Greece

Comparison of Systematic and Random Grammar-Based Test suite Construction
SASUF 2019 Workshop

Stellenbosch

Spectrum-Based Fault Localization for Context-Free Grammars
AC21 Workshop (Co-located with ICTAC 2018 Conference)

Stellenbosch

Research Tools and Prototypes

lexinfer

A prototypical software that learns the structure of the input space of black-box systems from a set of positive training examples. It found bugs and inconsistencies in popular open-source systems where these systems do not correctly implement the intended specification.

idefuzz

An LLVM-based patch testing framework that leverages two powerful automated test generation techniques to identify bad fixes. Successfully confirmed bad fixes in popular C open-source libraries.

gfixr

gfixr is a fully automatic grammar repair tool. It takes as input a faulty grammar and test suite, applies small-scale transformations and gives as output the repaired grammar variant that is consistent with the input tests and an optional membership oracle for the target language.

gtestr

gtestr is a grammar-based test-suite generation tool that can read a context-free grammar in EBNF notation and produce from this a wide range of test suites with different properties. In particular, **gtestr** can produce positive test suites that ensure that a variety of different grammar coverage criteria are satisfied, as well as negative test suites that ensure that the tests have a single, well-defined syntax error.

Teaching Activities

All teaching experience described here was gained in my role as a Teaching Assistant.

Software Engineering II

Humboldt-Universität zu Berlin, Germany

- **Lecture:** Symbolic Execution, Concolic Execution and Hybrid Fuzzing.

- **Semester:** Summer 2025

Compiler Construction

Humboldt-Universität zu Berlin, Germany

- **Lecture:** Grammar-based testing

- **Semester:** Summer 2025

Compiler Construction

Stellenbosch University, South Africa

- July 2019 - November 2019

- July 2020 - November 2020

- February 2021 - June 2021

- February 2022 - June 2022

Introduction to Programming I

Stellenbosch University, South Africa

- February 2019 - May 2019

- February 2020 - May 2020

- March 2021 - June 2021

Introduction to Programming II

Stellenbosch University, South Africa

- July 2019 - November 2019

Students

MSc students:

1. Hasan Karam, Humboldt-Universität zu Berlin (*Graduated*). Repairing Inferred Context-Free Grammars Learned in Black-Box Settings
2. Gareth Dalton, Stellenbosch University (*Current*). Learning the lexical structure of software systems.

BSc Students:

1. Vladislav Yudin, Humboldt-Universität zu Berlin (*Graduated*). Automated Validation of Patch Correctness and Maintainability with Symbolic Execution.

Service

PC Member International Conference on Software Engineering (ICSE 2025),

International Conference on Software Language Engineering (SLE 2025)

External Reviewer International Conference on Software Engineering (ICSE 2024, 2026),

International Conference on Automated Software Engineering (ASE 2024, 2026),

International Conference on Foundations of Software Engineering (FSE 2024, 2026),

International Conference on Fundamental Approaches to Software Engineering (FASE 2026)

Artefact Reviewer International Conference on Software Language Engineering (SLE 2021)

Awards, Honours and Achievements

- o ACM SIGPLAN Distinguished Paper Award, 2019.
- o ACM travel support and 100% fee waiver for SPLASH, 2019.

Additional Information

- o **Programming languages and tools:** Java, Python, C/C++, Haskell, Kotlin, Javascript, Sqlite, PostgreSQL, MySQL, Git, Docker and Maven.
- o **Languages:** Sesotho (native), English (fluent).
- o **Citizenship:** Lesotho. **Residence:** Germany

References

Prof. Bernd Fischer

PhD Supervisor

Prof. Lars Grunske

Mentor

Prof. Konstantinos Sagonas

Co-author

Prof. Eric Van Wyk

PhD Examiner

Stellenbosch University, South Africa

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Uppsala University, Sweden

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University of Minnesota, USA

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